

12_Metrics_Dashboard_and_Audit_Framework

Project: Kristal Farms — Labrador Coast Partner Package

First target: Nain, Labrador

Expansion logic: Replicable Labrador coastal communities

Document type: Partner-facing due diligence appendix

Status: Draft for partner review

1. Purpose

This document defines the metrics, dashboard structure, reporting cadence, audit process, and data-governance rules for the Kristal Farms Labrador coast project.

The purpose is to make the project measurable from the beginning. Partners, community representatives, technical reviewers, and funders should be able to see whether the project is delivering on its central claims:

1. clean hydro-powered compute;
2. efficient cooling;
3. useful heat delivery to the host community;
4. reduced diesel dependence;
5. reliable fibre-connected compute operations;
6. local jobs and training;
7. transparent community benefit reporting; and
8. privacy-preserving black-box tenancy.

The dashboard is not a marketing tool. It is a project-accountability tool.

2. Core dashboard principle

Kristal Farms should use a **small, stable, auditable metric set**.

The project should avoid constantly changing metrics, adding speculative indicators, or publishing numbers that cannot be measured consistently. Every public-facing metric should have:

- a clear definition;
- a measurement source;
- a reporting frequency;
- an accountable owner;
- a validation method;
- a status: confirmed, estimated, or under validation.

The project should report enough to prove performance, without exposing tenant data, private community information, or commercially sensitive details.

3. Dashboard categories

The dashboard should be organized into six categories:

1. Energy and compute efficiency
2. Heat recycling and diesel displacement
3. Water and environmental performance
4. Connectivity and pad operations
5. Community benefits
6. Governance, audit, and incidents

Each category should include a small number of headline metrics, supported by more detailed internal logs.

4. Energy and compute efficiency metrics

4.1 Power Usage Effectiveness — PUE

Definition:

PUE measures the ratio of total facility power to IT equipment power.

$$\text{PUE} = \text{Total facility energy} / \text{IT equipment energy}$$

Purpose:

PUE shows how much non-IT overhead the site requires. A lower PUE indicates that more energy is going directly into compute rather than cooling, pumping, lighting, auxiliary loads, or site overhead.

Report as:

Monthly PUE

Quarterly PUE

Annual average PUE

Winter PUE

Summer PUE

Measurement source:

- main site energy meter;
- pad-level IT meters;
- auxiliary load meters for pumps, dry coolers, heat pumps, network equipment, site lighting, and control systems.

Public dashboard status:

Public, aggregated.

Notes:

Do not publish a final target until the site design and seasonal heat/cooling model are validated. The dashboard can show a placeholder target range during pre-feasibility, then a binding target after technical design.

4.2 IT energy consumed

Definition:

Total electrical energy consumed by tenant compute equipment.

Report as:

kWh_IT per month

MWh_IT per quarter

MWh_IT per year

Measurement source:

- pad-level metered power handoff;
- tenant container energy meter;
- site supervisory control system.

Public dashboard status:

Public, aggregated across pads. Do not report tenant-specific values unless permitted by contract.

4.3 Facility overhead energy

Definition:

Energy used by non-IT systems required to operate the site.

Examples include:

- pumps;
- heat exchangers and controls;
- heat pump booster, where used;
- dry coolers;

- network and NOC equipment;
- lighting;
- security systems;
- site controls.

Report as:

kWh_auxiliary per month

% of total facility energy

Public dashboard status:

Public, aggregated.

4.4 Renewable energy share

Definition:

Percentage of site energy supplied by local hydro or other confirmed renewable sources.

Report as:

% renewable energy share

MWh renewable energy supplied

MWh backup/diesel energy used, if any

Measurement source:

- hydro plant meter;
- substation meter;
- diesel generator runtime and fuel logs;
- backup energy meter, if applicable.

Public dashboard status:

Public.

Notes:

This metric should distinguish normal operations from emergency backup events.

5. Heat recycling and diesel displacement metrics

5.1 Useful heat delivered

Definition:

Thermal energy delivered to productive local uses.

Report as:

MWh_th delivered to public buildings

MWh_th delivered to homes

MWh_th delivered to greenhouse

MWh_th delivered to thermal storage

Total MWh_th useful heat delivered

Measurement source:

- heat meters at building substations;
- greenhouse heat meter;
- thermal storage charge/discharge meters;
- central heat loop sensors.

Public dashboard status:

Public, aggregated. Building-by-building reporting should require community approval.

5.2 Heat Utilization Factor — HUF

Definition:

HUF measures the share of available server waste heat that is productively reused.

$$\text{HUF} = \text{Useful heat reused} / \text{Total recoverable server heat}$$

Purpose:

HUF is the central metric for the heat-first model. It shows whether the project is actually turning compute waste heat into community value.

Report as:

Monthly HUF

Winter HUF

Summer HUF

Annual average HUF

HUF by heat sink category

Measurement source:

- IT loop temperature and flow sensors;
- building loop heat meters;
- greenhouse heat meters;
- thermal storage meters;
- reject-loop meters.

Public dashboard status:

Public.

Notes:

Seasonal differences should be expected. Winter HUF should normally be higher because heating demand is higher. Summer HUF depends on greenhouse demand, storage capacity, domestic hot water demand, and permitted rejection limits.

5.3 Heat rejected

Definition:

Thermal energy that is not reused or stored and must be rejected to a permitted sink.

Report as:

MWh_{th} rejected to air-side dry coolers

MWh_{th} rejected to non-contact bay/seawater exchanger

% of available heat rejected

Measurement source:

- reject-loop heat meters;
- dry cooler control logs;
- cold-source intake/discharge temperature sensors;
- environmental compliance records.

Public dashboard status:

Public, aggregated.

Notes:

The operating hierarchy is:

Reuse → Store → Reject

Rejection should be reported transparently, but it should not be framed as failure if it occurs within permitted limits and after reuse/storage capacity has been exhausted.

5.4 Diesel avoided

Definition:

Estimated diesel fuel not burned because server heat replaced heating fuel and hydro-backed operations reduced diesel use.

Report as:

Litres diesel avoided

MWh equivalent avoided

tCO₂e avoided, if emission factor is approved

Measurement source:

- baseline heating fuel records;
- building heat meters;
- diesel generator fuel logs;
- approved diesel-to-energy conversion factor;
- approved emissions factor, if used.

Public dashboard status:

Public, aggregated.

Notes:

The baseline must be agreed before public claims are made. The metric should distinguish:

1. Heating diesel avoided
2. Electricity-generation diesel avoided
3. Emergency diesel still used

5.5 Heat priority delivery

Definition:

Tracks whether heat is delivered according to the agreed community priority order.

Recommended priority order:

1. Critical public buildings
2. Homes / residential heating
3. Domestic hot water
4. Greenhouse and food systems
5. Thermal storage
6. Rejection

Measurement source:

- heat dispatch logs;
- building substation meters;
- Heat Committee review records;
- incident logs.

Public dashboard status:

Public summary only.

6. Water and environmental performance metrics

6.1 Water Usage Effectiveness — WUE

Definition:

WUE measures water consumed for cooling per unit of IT energy.

WUE = Litres of water consumed / kWh_IT

Purpose:

WUE confirms whether the project is avoiding evaporative cooling and large water consumption.

Expected design logic:

Kristal Farms uses closed loops and non-contact heat exchange. Therefore, water consumption should be near zero aside from maintenance, treatment, or minor top-up requirements.

Report as:

Litres consumed per month

Litres/kWh_IT

Maintenance top-up volume

Public dashboard status:

Public.

6.2 Cold-source temperature compliance

Definition:

Tracks intake and discharge temperatures at the non-contact cold-source exchanger, where applicable.

Report as:

Intake temperature

Discharge temperature

Delta-T

Number of exceedance events

Duration of exceedance events

Measurement source:

- temperature sensors;
- environmental monitoring logs;
- cold-source pump and exchanger records.

Public dashboard status:

Public summary. Detailed raw logs may be provided to regulators or auditors.

6.3 Environmental incidents

Definition:

Any reportable incident related to cooling, spills, wildlife, water temperature, fuel handling, construction, waste, noise, or permit compliance.

Report as:

Number of incidents

Severity level

Corrective action status

Days to close

Public dashboard status:

Public summary, subject to legal and regulatory requirements.

7. Connectivity and pad operations metrics

7.1 Fibre availability

Definition:

Percentage of time fibre connectivity is available to support tenant operations.

Report as:

Monthly fibre availability %

Quarterly fibre availability %

Number of outage events

Total outage minutes

Measurement source:

- network monitoring system;
- NOC logs;
- telecom partner reports.

Public dashboard status:

Public, aggregated. Do not publish tenant traffic content or sensitive network metadata.

7.2 Latency p95

Definition:

The 95th percentile network latency over a defined measurement path.

Report as:

Latency p95 to agreed exchange point

Latency p95 to agreed cloud/tenant handoff point

Monthly average and peak periods

Measurement source:

- NOC monitoring;
- synthetic network tests;
- telecom partner reports.

Public dashboard status:

Public if the path is not commercially sensitive. Otherwise, NDA-only.

7.3 Pad uptime

Definition:

Percentage of time each compute pad has available power, cooling interface, and fibre handoff.

Report as:

Pad uptime %

Power availability %

Cooling-interface availability %

Fibre-handoff availability %

Measurement source:

- pad meter;
- cooling interface sensors;
- NOC logs;
- site control system.

Public dashboard status:

Public aggregated. Tenant-specific pad reporting should follow lease terms.

7.4 Pad occupancy

Definition:

Share of available pad capacity that is leased, installed, or actively used.

Recommended sub-metrics:

Leased capacity

Installed capacity

Active IT load

Available expansion capacity

Measurement source:

- lease records;
- pad meters;
- tenant activation records;
- site capacity register.

Public dashboard status:

Public aggregated. Tenant identities and commercial details remain confidential unless disclosed by agreement.

8. Community benefit metrics

8.1 Local jobs

Definition:

Number of jobs created or supported by the project, with priority given to local and community-based employment.

Report as:

Direct local jobs

Operations jobs

Construction jobs

Training placements

Seasonal jobs

Measurement source:

- payroll records;
- contractor reports;
- community employment reporting;
- training program records.

Public dashboard status:

Public aggregated. No personal employee information.

8.2 Training hours

Definition:

Total project-funded or project-supported training hours delivered to local participants.

Report as:

Training hours completed

Number of participants

Certifications achieved

Apprenticeships or placements created

Measurement source:

- training provider records;
- attendance logs;
- certification records;
- project workforce reports.

Public dashboard status:

Public aggregated.

8.3 Local procurement

Definition:

Share of project purchasing directed to local or regional suppliers where feasible.

Report as:

Local procurement spend

Regional procurement spend

Total procurement spend

% local/regional procurement

Measurement source:

- procurement records;
- contractor reporting;
- invoice records.

Public dashboard status:

Public aggregated. Supplier-level details may be NDA-only.

8.4 Greenhouse output

Definition:

Food or plant output from heat-supported greenhouse operations.

Report as:

kg produce grown

kg produce distributed locally

greenhouse operating days

MWh_{th} used by greenhouse

Measurement source:

- greenhouse records;
- heat meters;
- distribution records;
- community food program records.

Public dashboard status:

Public.

Notes:

The greenhouse metric should be treated as a community-value metric, not as a guaranteed commercial agriculture claim until greenhouse design, operator, and growing plan are confirmed.

8.5 Community benefits delivered**Definition:**

Tracks benefits delivered under any Community Benefits Agreement, Impact Benefit Agreement, or equivalent project agreement.

Possible categories:

heat credits

training commitments

local hiring commitments

community fund contributions

public building heat support

connectivity improvements

education or knowledge access programs

Measurement source:

- agreement obligations register;
- project council minutes;
- community benefit reports;

- finance records.

Public dashboard status:

Public summary, subject to agreement terms.

9. Governance and trust metrics

9.1 Dashboard publication cadence

Minimum cadence:

Monthly dashboard update

Quarterly performance review

Annual public report

Annual independent audit

The monthly dashboard should be short and readable. The quarterly review should be detailed enough for committees and partners. The annual report should summarize technical, environmental, community, and governance performance.

9.2 Committee review status

Definition:

Tracks whether required reviews occurred.

Report as:

Project Council review completed: yes/no

Heat Committee review completed: yes/no

Environment Committee review completed: yes/no

Community reporting session completed: yes/no

Public dashboard status:

Public.

Notes:

Committee names and final structure may change after community process. The dashboard should adapt to the governance structure agreed with the community.

9.3 Grievances and dispute resolution

Definition:

Tracks formal grievances submitted, reviewed, resolved, or escalated.

Report as:

Number of grievances received

Number resolved

Number open

Average days to resolution

Escalated cases

Public dashboard status:

Public aggregated. No personal or confidential details.

9.4 Corrective action closure

Definition:

Tracks whether corrective actions from audits, incidents, or committee reviews are closed on time.

Report as:

Corrective actions opened

Corrective actions closed

Corrective actions overdue

Average closure time

Public dashboard status:

Public summary.

10. Black-box tenancy and data boundaries

The dashboard must respect the black-box tenancy model.

10.1 Host-visible data

The host may monitor and report physical infrastructure metrics, including:

energy use

power availability

cooling loop temperature

cooling loop flow

cooling delta-T

pad uptime

bandwidth volume

fibre availability

technical alarms

heat delivery

water and environmental compliance metrics

10.2 Host-excluded data

The host must not access or publish:

tenant application logs

tenant data

tenant model content

tenant prompts or outputs

tenant packet payloads

sensitive tenant metadata

user-level traffic details

proprietary workload details

10.3 Reporting rule

All public dashboard information should be aggregated unless a tenant, community body, or partner has explicitly agreed to disclose more detail.

11. Data architecture

11.1 Measurement layers

The monitoring system should use four layers:

1. Sensor layer

- meters, flow sensors, temperature sensors, network monitors, environmental sensors

2. Site control layer

- SCADA or equivalent system for power, cooling, heat, and alarms

3. Dashboard data layer

- cleaned, aggregated, non-sensitive data for public and partner reporting

4. Audit archive

- tamper-evident logs, raw meter data, incident records, and signed reports

11.2 Raw data retention

Physical infrastructure logs may be retained long term for trend analysis, auditability, and replication learning. Retention should be documented in the Data Room and should exclude tenant data.

11.3 Data quality status labels

Every metric should be labeled as one of the following:

Measured

Estimated

Modeled

Under validation

Not yet available

This avoids presenting early estimates as final performance.

12. Reporting cadence

12.1 Monthly dashboard

The monthly dashboard should include:

PUE

WUE

MWh_th useful heat delivered

HUF

diesel avoided

fibre availability

pad uptime

pad occupancy

local jobs

training hours

greenhouse output

open incidents

open corrective actions

12.2 Quarterly review

The quarterly review should include:

dashboard trend analysis

seasonal heat performance

fibre and uptime review

incident review

community benefit tracking

risk updates

committee recommendations

corrective action status

12.3 Annual public report

The annual report should include:

annual metric summary

comparison with agreed targets

community benefit summary

environmental performance summary

financial categories without sensitive details

governance and grievance summary

auditor statement

next-year improvement plan

12.4 Independent audit

The independent audit should review:

meter accuracy

PUE/WUE calculation method

heat metering and HUF calculation

fuel displacement baseline

fibre and uptime records

environmental compliance logs

community benefit commitments

privacy/data-boundary compliance

corrective action closure

13. Target-setting approach

Targets should be set in phases.

Phase 0 — Concept and partner alignment

Use provisional metrics only.

Status: modeled / under validation

No binding public performance claims

Phase 1 — Site confirmation

Confirm what can be measured.

Metering plan

baseline fuel data request

tenant privacy boundary

heat sink inventory

fibre baseline

Phase 2 — Pre-feasibility

Define draft targets.

draft PUE target

draft WUE target

draft HUF seasonal range

draft uptime target

draft diesel-avoidance baseline

Phase 3 — Pilot pad

Switch from modeled to measured values.

measured pad energy

measured heat delivery

measured reject-loop behavior

measured fibre performance

measured auxiliary load

Phase 4 — Expansion pads

Set binding operational targets.

annual performance target

heat allocation target

community benefit target

fibre SLA target

audit requirements

14. Example dashboard layout

Kristal Farms Labrador Coast Dashboard

Reporting month: [Month Year]

Site: Nain pilot / Labrador coast replication

Status: Concept / Pre-feasibility / Pilot / Operating

Energy and compute

- PUE: [value/status]

- IT energy: [MWh_IT]

- Renewable energy share: [%]

Heat and diesel displacement

- Useful heat delivered: [MWh_th]

- HUF: [%]

- Heat rejected: [MWh_th]

- Diesel avoided: [litres / MWh / tCO_{2e}]

Water and environment

- WUE: [L/kWh_IT]
- Cold-source delta-T: [°C]
- Environmental incidents: [number/status]

Connectivity and operations

- Fibre availability: [%]
- Latency p95: [ms]
- Pad uptime: [%]
- Pad occupancy: [%]

Community benefits

- Local jobs: [number]
- Training hours: [hours]
- Greenhouse output: [kg]
- Community benefits delivered: [summary]

Governance and audit

- Committee reviews completed: [status]
 - Open grievances: [number]
 - Open corrective actions: [number]
 - Next public report: [date]
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15. Partner decision points

Partners should confirm the following before pilot deployment:

1. Which metrics are public, NDA-only, or internal.
 2. Which entity owns each data source.
 3. Which meters and sensors are revenue-grade versus operational-grade.
 4. Which baselines are accepted for diesel avoided.
 5. Which latency/fibre paths are used for SLA reporting.
 6. Which community benefits are binding commitments.
 7. Which committee or council reviews the dashboard each quarter.
 8. Which independent auditor or technical reviewer validates annual results.
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16. Metrics register

Category	Metric	Unit	Public?	Status before pilot	Primary owner
Energy	PUE	ratio	Yes	Modeled	Site operator
Energy	IT energy	MWh_IT	Aggregated	Metering plan	Site operator / tenant interface
Energy	Renewable share	%	Yes	Under validation	Utility / site operator

Category	Metric	Unit	Public?	Status before pilot	Primary owner
Heat	Useful heat delivered	MWh_th	Yes	Heat-sink inventory	Heat operator
Heat	HUF	%	Yes	Modeled	Heat operator
Heat	Heat rejected	MWh_th	Yes	Modeled	Heat operator
Diesel	Diesel avoided	L / MWh / tCO _{2e}	Yes	Baseline needed	Site operator / community
Water	WUE	L/kWh_IT	Yes	Design estimate	Site operator
Environment	Cold-source delta-T	°C	Yes summary	Permitting needed	Environment lead
Fibre	Fibre availability	%	Yes aggregate	Telecom design	Telecom/NOC
Fibre	Latency p95	ms	Conditional	Telecom design	Telecom/NOC
Operations	Pad uptime	%	Yes aggregate	SLA draft	Site operator
Operations	Pad occupancy	%	Yes aggregate	Commercial pipeline	Site operator
Community	Local jobs	count	Yes aggregate	Workforce plan	Project council / operator
Community	Training hours	hours	Yes aggregate	Training plan	Project council / operator

Category	Metric	Unit	Public?	Status before pilot	Primary owner
Food	Greenhouse output	kg	Yes	Greenhouse plan	Greenhouse operator
Governance	Grievances	count/status	Yes aggregate	Process needed	Project council
Audit	Corrective actions	count/status	Yes summary	Audit plan	Site operator / auditor

17. What should not be reported publicly

The public dashboard should not include:

tenant names without consent

tenant application data

tenant logs

tenant model information

packet payloads

sensitive network metadata

employee names or personal records

household-level heat consumption without consent

commercial pricing

unvalidated financial projections

security-sensitive site details

18. Open design items

The following items should be finalized during partner alignment and community process:

1. Final metric list for the first public dashboard.
2. Exact PUE, WUE, HUF, uptime, and fibre targets.
3. Diesel baseline method and emissions factor.
4. Heat allocation priority order.
5. Greenhouse measurement method and operator responsibility.
6. Committee review structure.
7. Auditor selection and audit scope.
8. Data retention policy for physical infrastructure logs.
9. Public/NDA/internal classification for each metric.
10. Dashboard publication platform and update workflow.

19. Summary

The Metrics Dashboard and Audit Framework makes Kristal Farms measurable and accountable.

The most important metrics are:

PUE

WUE

Heat Utilization Factor

MWh_th delivered

Diesel avoided

Fibre availability

Latency p95

Pad uptime

Pad occupancy

Local jobs

Training hours

Greenhouse output

Community benefits delivered

Incident and corrective-action closure

The dashboard should prove the central project thesis: Labrador coast compute infrastructure should not only consume clean power; it should also deliver local heat, reduce diesel dependence, improve connectivity, create skills, and maintain transparent governance while protecting tenant confidentiality.